Question 1 — From Equations to Matrix Form (10 Points)

Consider the following system of four equations in four unknowns:

$$
\begin{aligned}
2x_1 - x_2 + 0x_3 + 3x_4 &= 7 \\
-4x_1 + 5x_2 + x_3 - 2x_4 &= -6 \\
x_1 + 2x_2 + 4x_3 + 0x_4 &= 10 \\
0x_1 - 3x_2 + 2x_3 + x_4 &= 1
\end{aligned}
$$

1. Write the system in matrix form $\mathbf{A}\mathbf{x} = \mathbf{b}$.
2. Clearly identify $\mathbf{A}$, $\mathbf{x}$, and $\mathbf{b}$.
3. State the dimensions of $\mathbf{A}$, $\mathbf{x}$, and $\mathbf{b}$.

Solution for Question 1

1. Write the system in matrix form $\mathbf{A}\mathbf{x} = \mathbf{b}$.

$$
\begin{bmatrix} 2 & -1 & 0 & 3 \\ -4 & 5 & 1 & -2 \\ 1 & 2 & 4 & 0 \\ 0 & -3 & 2 & 1 \end{bmatrix}
\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}
=
\begin{bmatrix} 7 \\ -6 \\ 10 \\ 1 \end{bmatrix}
$$

2. Clearly identify $\mathbf{A}$, $\mathbf{x}$, and $\mathbf{b}$.

$$
\mathbf{A} = \begin{bmatrix} 2 & -1 & 0 & 3 \\ -4 & 5 & 1 & -2 \\ 1 & 2 & 4 & 0 \\ 0 & -3 & 2 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 7 \\ -6 \\ 10 \\ 1 \end{bmatrix}
$$

3. State the dimensions of $\mathbf{A}$, $\mathbf{x}$, and $\mathbf{b}$.

- $\mathbf{A}$ is 4×4
- $\mathbf{x}$ is 4×1
- $\mathbf{b}$ is 4×1

Question 2 — 2D Linear Transformations (10 Points)

Let the vector

$$
\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}
$$

Consider the following three transformation matrices:

$$
\mathbf{T}_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \qquad \mathbf{T}_2 = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix} \qquad \mathbf{T}_3 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}
$$

1. Compute $\mathbf{y}_k = \mathbf{T}_k \mathbf{x}$ for $k = 1, 2, 3$.
2. On a single set of axes, sketch the original vector $\mathbf{x}$ and the three transformed vectors $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$.

Solution for Question 2

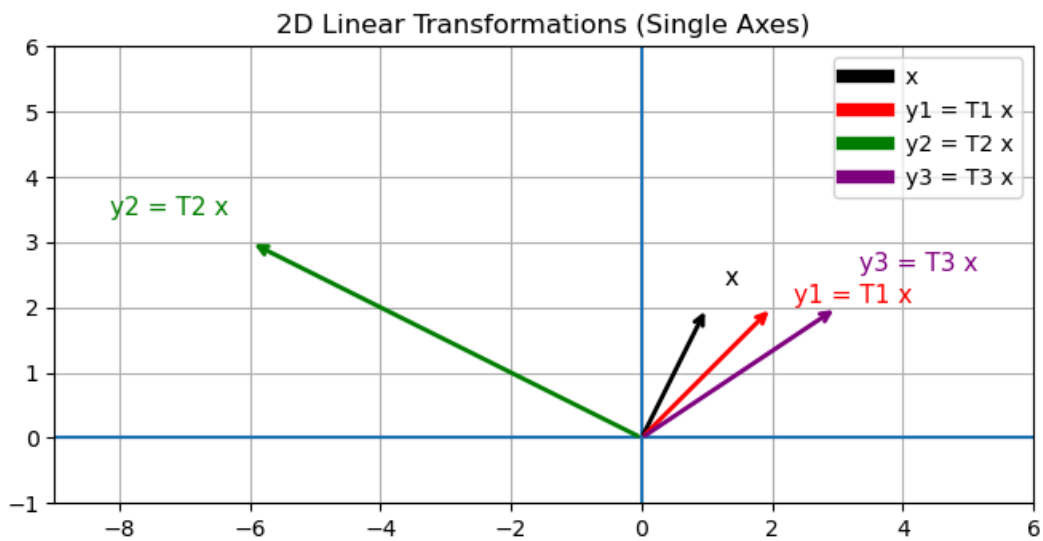
1. Compute $\mathbf{y}_k = \mathbf{T}_k\mathbf{x}$ for $k = 1, 2, 3$.

$$\mathbf{y}_1 = \mathbf{T}_1\mathbf{x} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$\mathbf{y}_2 = \mathbf{T}_2\mathbf{x} = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -6 \\ 3 \end{bmatrix}$$

$$\mathbf{y}_3 = \mathbf{T}_3\mathbf{x} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

2. On a single set of axes, sketch the original vector $\mathbf{x}$ and the three transformed vectors $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$.



Question 3 — Matrix Multiplication (10 Points)

Let

$$\mathbf{A} = \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ 4 & 1 \\ -1 & 2 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 2 & 0 & -1 & 3 & 1 \\ -2 & 4 & 5 & 0 & -3 \end{bmatrix}$$

1. State the dimensions of $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{AB}$.
2. Compute the product $\mathbf{C} = \mathbf{AB}$.
3. Show your work clearly for **two entries** of $\mathbf{C}$:
   - c_{11}
   - c_{34}

Solution for Question 3

1. State the dimensions of $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{AB}$.

- $\mathbf{A}$ is 4×2
- $\mathbf{B}$ is 2×5
- $\mathbf{AB}$ is 4×5

2. Compute the product $\mathbf{C} = \mathbf{AB}$.

$$\mathbf{C} = \mathbf{AB} = \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ 4 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 & -1 & 3 & 1 \\ -2 & 4 & 5 & 0 & -3 \end{bmatrix} = \begin{bmatrix} 6 & -8 & -11 & 3 & 7 \\ -6 & 12 & 15 & 0 & -9 \\ 6 & 4 & 1 & 12 & 1 \\ -6 & 8 & 11 & -3 & 7 \end{bmatrix}$$

3. Show your work clearly for **two entries** of $\mathbf{C}$:
   - $c_{11} = 1 * 2 + (-2) * (-2) = 6$
   - $c_{34} = 4 * 3 + 1 * 0 = 12$

Question 4 — Block Matrix Multiplication (10 Points)

Consider the matrices

$$\mathbf{K} \in \mathbb{R}^{6\times 6} \qquad \mathbf{X} \in \mathbb{R}^{6\times 3}$$

Matrix $\mathbf{K}$ is partitioned into four blocks:

- $\mathbf{K}_{11} \in \mathbb{R}^{4\times 4}$
- $\mathbf{K}_{12} \in \mathbb{R}^{4\times 2}$
- $\mathbf{K}_{21} \in \mathbb{R}^{2\times 4}$
- $\mathbf{K}_{22} \in \mathbb{R}^{2\times 2}$

and matrix $\mathbf{X}$ is partitioned compatibly as:

- $\mathbf{X}_1 \in \mathbb{R}^{4\times 3}$
- $\mathbf{X}_2 \in \mathbb{R}^{2\times 3}$

1. Sketch out the partitioning on the matrices (you dont need to write out all the entries). Write the **partitioned forms** of $\mathbf{K}$ and $\mathbf{X}$ using block notation,
2. Write the block multiplication expression for

$$\mathbf{Y} = \mathbf{KX}$$

   in terms of the submatrices.
3. For each product appearing in your expression, state its dimensions. For example:

$$\mathbf{K}_{12}\mathbf{X}_2 : \quad (4\times 2)(2\times 3) = (4\times 3)$$

4. State the final dimension of $\mathbf{Y}$.

You do **not** need to perform any numerical multiplication — the goal is to correctly set up the block multiplication and verify dimensional consistency.

Solution for Question 4

1. Sketch out the partitioning on the matrices (you dont need to write out all the entries). Write the **partitioned forms** of $\mathbf{K}$ and $\mathbf{X}$ using block notation,

$$\mathbf{K} = \left[\begin{array}{cccc|cc} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ \hline k_{51} & k_{52} & k_{53} & k_{54} & k_{55} & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{array}\right] = \begin{bmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} \\ \mathbf{K}_{21} & \mathbf{K}_{22} \end{bmatrix}, \quad \mathbf{K}_{11} \in \mathbb{R}^{4\times 4},\ \mathbf{K}_{12} \in \mathbb{R}^{4\times 2},\ \mathbf{K}_{21} \in \mathbb{R}^{2\times 4},\ \mathbf{K}_{22} \in \mathbb{R}^{2\times 2}.$$

$$\mathbf{X} = \left[\begin{array}{ccc} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \\ x_{41} & x_{42} & x_{43} \\ \hline x_{51} & x_{52} & x_{53} \\ x_{61} & x_{62} & x_{63} \end{array}\right] = \mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix}, \quad \mathbf{X}_1 \in \mathbb{R}^{4\times 3},\ \mathbf{X}_2 \in \mathbb{R}^{2\times 3}.$$

2. Write the block multiplication expression for

$$\mathbf{Y} = \mathbf{K}\mathbf{X}$$

in terms of the submatrices.

$$\mathbf{Y} = \mathbf{K}\mathbf{X} = \begin{bmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} \\ \mathbf{K}_{21} & \mathbf{K}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{K}_{11}\mathbf{X}_1 + \mathbf{K}_{12}\mathbf{X}_2 \\ \mathbf{K}_{21}\mathbf{X}_1 + \mathbf{K}_{22}\mathbf{X}_2 \end{bmatrix}.$$

3. For each product appearing in your expression, state its dimensions.

$$\begin{aligned} \mathbf{K}_{11}\mathbf{X}_1 : &\quad (4\times 4)(4\times 3) = (4\times 3) \\ \mathbf{K}_{12}\mathbf{X}_2 : &\quad (4\times 2)(2\times 3) = (4\times 3) \\ \mathbf{K}_{21}\mathbf{X}_1 : &\quad (2\times 4)(4\times 3) = (2\times 3) \\ \mathbf{K}_{22}\mathbf{X}_2 : &\quad (2\times 2)(2\times 3) = (2\times 3) \end{aligned}$$

4. State the final dimension of $\mathbf{Y}$.

$$\mathbf{Y} : \quad (6\times 6)(6\times 3) = (6\times 3)$$

Question 5 — Gauss–Jordan (10 Points)

Let

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 & 1 \\ 1 & 3 & 1 & 0 \\ 0 & 2 & 4 & 1 \\ 1 & 0 & 1 & 2 \end{bmatrix} \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

1. Form the augmented matrix $[\mathbf{A} \mid \mathbf{I}]$.
2. Perform Gauss–Jordan elimination **only through the first two pivots** (i.e., make the first two pivot columns match $\mathbf{I}$).
3. Clearly show your row operations and the partially reduced matrix.

Solution for Question 5

1. Form the augmented matrix $[\mathbf{A} \mid \mathbf{I}]$.

$$[\mathbf{A}|\mathbf{I}] = \left[\begin{array}{cccc|cccc} 2 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 3 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right]$$

2. Perform Gauss–Jordan elimination **only through the first two pivots** (i.e., make the first two pivot columns match $\mathbf{I}$).

3. Clearly show your row operations and the partially reduced matrix.

Pivot 1 (column 1): Row operation:

- $R_1 \leftarrow \frac{1}{2}R_1$

$$\left[\begin{array}{cccc|cccc} 1 & 0.5 & 0 & 0.5 & 0.5 & 0 & 0 & 0 \\ 1 & 3 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right]$$

- $R_2 \leftarrow R_2 - R_1$
- $R_4 \leftarrow R_4 - R_1$

$$\left[\begin{array}{cccc|cccc} 1 & 0.5 & 0 & 0.5 & 0.5 & 0 & 0 & 0 \\ 0 & 2.5 & 1 & -0.5 & -0.5 & 1 & 0 & 0 \\ 0 & 2 & 4 & 1 & 0 & 0 & 1 & 0 \\ 0 & -0.5 & 1 & 1.5 & -0.5 & 0 & 0 & 1 \end{array}\right]$$

Pivot 2 (column 2):

- $R_2 \leftarrow \frac{1}{2.5}R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 0.5 & 0 & 0.5 & 0.5 & 0 & 0 & 0 \\ 0 & 1 & 0.4 & -0.2 & -0.2 & 0.4 & 0 & 0 \\ 0 & 2 & 4 & 1 & 0 & 0 & 1 & 0 \\ 0 & -0.5 & 1 & 1.5 & -0.5 & 0 & 0 & 1 \end{array}\right]$$

- $R_1 \leftarrow R_1 - \frac{1}{2}R_2$
- $R_3 \leftarrow R_3 - 2R_2$
- $R_4 \leftarrow R_4 + \frac{1}{2}R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -0.2 & 0.6 & 0.6 & -0.2 & 0 & 0 \\ 0 & 1 & 0.4 & -0.2 & -0.2 & 0.4 & 0 & 0 \\ 0 & 0 & 3.2 & 1.4 & 0.4 & -0.8 & 1 & 0 \\ 0 & 0 & 1.2 & 1.4 & -0.6 & 0.2 & 0 & 1 \end{array}\right]$$

Question 6 — LU Decomposition (10 Points)

Using the same matrix $\mathbf{A}$ from Question 5:

1. Perform Gaussian elimination (no pivoting, i.e., swapping rows) to produce $\mathbf{U}$.
2. Record the elimination multipliers to construct $\mathbf{L}$ (unit diagonal).
3. Verify your result by computing (by hand) the product $\mathbf{LU}$ and checking that it equals $\mathbf{A}$.

Solution for Question 6

1. Perform Gaussian elimination (no pivoting, i.e., swapping rows) to produce $\mathbf{U}$.

Ccolumn 1:

Elimination multipliers

$$\ell_{21} = \frac{a_{21}}{a_{11}} = \frac{1}{2}, \qquad \ell_{31} = \frac{a_{31}}{a_{11}} = 0, \qquad \ell_{41} = \frac{a_{41}}{a_{11}} = \frac{1}{2}$$

Row operations (update the matrix to start forming $\mathbf{U}$)

- $R_2 \leftarrow R_2 - \frac{1}{2}R_1$
- $R_4 \leftarrow R_4 - \frac{1}{2}R_1$

$$\begin{bmatrix} 2 & 1 & 0 & 1 \\ 0 & 2.5 & 1 & -0.5 \\ 0 & 2 & 4 & 1 \\ 0 & -0.5 & 1 & 1.5 \end{bmatrix}$$

Ccolumn 2:

Elimination multipliers

$$\ell_{32} = \frac{a'_{32}}{a'_{22}} = \frac{4}{5}, \qquad \ell_{42} = \frac{a'_{42}}{a'_{22}} = -\frac{1}{5}$$

Row operations (update the matrix to start forming $\mathbf{U}$)

- $R_3 \leftarrow R_3 - \frac{4}{5}R_2$
- $R_4 \leftarrow R_4 + \frac{1}{5}R_2$

$$\begin{bmatrix} 2 & 1 & 0 & 1 \\ 0 & 2.5 & 1 & -0.5 \\ 0 & 0 & 3.2 & 1.4 \\ 0 & 0 & 1.2 & 1.4 \end{bmatrix}$$

Ccolumn 3:

Elimination multipliers

$$\ell_{43} = \frac{a'_{43}}{a'_{33}} = \frac{3}{8}$$

Row operations (update the matrix to start forming $\mathbf{U}$)

- $R_4 \leftarrow R_4 - \frac{3}{8}R_3$

$$\begin{bmatrix} 2 & 1 & 0 & 1 \\ 0 & 2.5 & 1 & -0.5 \\ 0 & 0 & 3.2 & 1.4 \\ 0 & 0 & 0 & 0.875 \end{bmatrix}$$

2. Record the elimination multipliers to construct $\mathbf{L}$ (unit diagonal).

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \ell_{21} & 1 & 0 & 0 \\ \ell_{31} & \ell_{32} & 1 & 0 \\ \ell_{41} & \ell_{42} & \ell_{43} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.5 & 1 & 0 & 0 \\ 0 & 0.8 & 1 & 0 \\ 0.5 & -0.2 & 0.375 & 1 \end{bmatrix}$$

3. Verify your result by computing (by hand) the product $\mathbf{LU}$ and checking that it equals $\mathbf{A}$.

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.5 & 1 & 0 & 0 \\ 0 & 0.8 & 1 & 0 \\ 0.5 & -0.2 & 0.375 & 1 \end{bmatrix}, \qquad \mathbf{U} = \begin{bmatrix} 2 & 1 & 0 & 1 \\ 0 & 2.5 & 1 & -0.5 \\ 0 & 0 & 3.2 & 1.4 \\ 0 & 0 & 0 & 0.875 \end{bmatrix}.$$

Compute $\mathbf{LU}$ row-by-row is a linear combination of rows of $\mathbf{U}$:

- **Row 1**:

$$[1,0,0,0]\mathbf{U} = \mathrm{R}_1(\mathbf{U}) = [2,1,0,1].$$

- **Row 2**:

$$[0.5,1,0,0]\mathbf{U} = 0.5\,\mathrm{R}_1(\mathbf{U}) + \mathrm{R}_2(\mathbf{U})$$
$$= 0.5[2,1,0,1] + [0,2.5,1,-0.5] = [1,3,1,0].$$

- **Row 3**:

$$[0,0.8,1,0]\mathbf{U} = 0.8\,\mathrm{R}_2(\mathbf{U}) + \mathrm{R}_3(\mathbf{U})$$
$$= 0.8[0,2.5,1,-0.5] + [0,0,3.2,1.4] = [0,2,4,1].$$

- **Row 4**:

$$[0.5,-0.2,0.375,1]\mathbf{U} = 0.5\,\mathrm{R}_1(\mathbf{U}) - 0.2\,\mathrm{R}_2(\mathbf{U}) + 0.375\,\mathrm{R}_3(\mathbf{U}) + \mathrm{R}_4(\mathbf{U})$$
$$= 0.5[2,1,0,1] - 0.2[0,2.5,1,-0.5] + 0.375[0,0,3.2,1.4] + [0,0,0,0.875]$$
$$= [1,0,1,2].$$

Therefore,

$$\mathbf{LU} = \begin{bmatrix} 2 & 1 & 0 & 1 \\ 1 & 3 & 1 & 0 \\ 0 & 2 & 4 & 1 \\ 1 & 0 & 1 & 2 \end{bmatrix} = \mathbf{A}.$$

Question 7 — Cholesky Factorization (20 Points)

Let the symmetric positive definite matrix

$$\mathbf{K} = \begin{bmatrix} 4 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

1. Compute the Cholesky factorization $\mathbf{K} = \mathbf{L}\mathbf{L}^T$.
2. Solve $\mathbf{K}\mathbf{x} = \mathbf{b}$ for each right-hand side:

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad \mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \qquad \mathbf{b}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

For each solve, show:

- forward substitution $\mathbf{L}\hat{\mathbf{b}} = \mathbf{b}$
- back substitution $\mathbf{L}^T\mathbf{x} = \hat{\mathbf{b}}$

Solution for Question 7

1. Compute the Cholesky factorization $\mathbf{K} = \mathbf{L}\mathbf{L}^T$.

Let

$$\mathbf{L} = \begin{bmatrix} \ell_{11} & 0 & 0 & 0 \\ \ell_{21} & \ell_{22} & 0 & 0 \\ \ell_{31} & \ell_{32} & \ell_{33} & 0 \\ \ell_{41} & \ell_{42} & \ell_{43} & \ell_{44} \end{bmatrix}$$

Calculate using the Cholesky formula:

- **Column 1**:

$$\ell_{11} = \sqrt{k_{11}} = \sqrt{4} = 2, \qquad \ell_{21} = \frac{k_{21}}{\ell_{11}} = \frac{1}{2}$$

$$\ell_{31} = \frac{k_{31}}{\ell_{11}} = 0, \qquad \ell_{41} = \frac{k_{41}}{\ell_{11}} = 0$$

- **Column 2**:

$$\ell_{22} = \sqrt{k_{22} - \ell_{21}^2} = \sqrt{3 - \left(\frac{1}{2}\right)^2} = \sqrt{\frac{11}{4}} = \frac{\sqrt{11}}{2}$$

$$\ell_{32} = \frac{k_{32} - \ell_{31}\ell_{21}}{\ell_{22}} = \frac{1 - 0}{\sqrt{11}/2} = \frac{2}{\sqrt{11}} = \frac{2\sqrt{11}}{11}$$

$$\ell_{42} = \frac{k_{42} - \ell_{41}\ell_{21}}{\ell_{22}} = \frac{0 - 0}{\ell_{22}} = 0$$

- **Column 3**:

$$\ell_{33} = \sqrt{k_{33} - \ell_{31}^2 - \ell_{32}^2} = \sqrt{2 - 0 - \left(\frac{2\sqrt{11}}{11}\right)^2} = \sqrt{2 - \frac{4}{11}} = \sqrt{\frac{18}{11}} = \frac{3\sqrt{22}}{11}$$

$$\ell_{43} = \frac{k_{43} - \ell_{41}\ell_{31} - \ell_{42}\ell_{32}}{\ell_{33}} = \frac{1 - 0 - 0}{3\sqrt{22}/11} = \frac{11}{3\sqrt{22}} = \frac{\sqrt{22}}{6}$$

- **Column 4**:

$$\ell_{44} = \sqrt{k_{44} - \ell_{41}^2 - \ell_{42}^2 - \ell_{43}^2} = \sqrt{2 - 0 - 0 - \left(\frac{\sqrt{22}}{6}\right)^2} = \sqrt{2 - \frac{22}{36}} = \sqrt{\frac{25}{18}} = \frac{5\sqrt{2}}{6}$$

Therefore:

$$\mathbf{L} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{\sqrt{11}}{2} & 0 & 0 \\ 0 & \frac{2\sqrt{11}}{11} & \frac{3\sqrt{22}}{11} & 0 \\ 0 & 0 & \frac{\sqrt{22}}{6} & \frac{5\sqrt{2}}{6} \end{bmatrix}$$

and $\mathbf{K} = \mathbf{L}\mathbf{L}^T$.

2. Solve $\mathbf{K}\mathbf{x} = \mathbf{b}$ for each right-hand side:

(a) $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

Forward substitution: $\mathbf{L}\hat{\mathbf{b}} = \mathbf{b}_1$

Let:

$$\hat{\mathbf{b}} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

Then:

$$2y_1 = 1 \Rightarrow y_1 = \frac{1}{2}$$

$$\frac{1}{2}y_1 + \frac{\sqrt{11}}{2}y_2 = 0 \Rightarrow y_2 = -\frac{\sqrt{11}}{22}$$

$$\frac{2\sqrt{11}}{11}y_2 + \frac{3\sqrt{22}}{11}y_3 = 0 \Rightarrow y_3 = \frac{\sqrt{22}}{66}$$

$$\frac{\sqrt{22}}{6}y_3 + \frac{5\sqrt{2}}{6}y_4 = 0 \Rightarrow y_4 = -\frac{\sqrt{2}}{30}$$

$$\hat{\mathbf{b}}_1 = \begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{11}}{22} \\ \frac{\sqrt{22}}{66} \\ -\frac{\sqrt{2}}{30} \end{bmatrix}$$

Back substitution: $\mathbf{L}^T\mathbf{x} = \hat{\mathbf{b}}_1$

$$\frac{5\sqrt{2}}{6}x_4 = -\frac{\sqrt{2}}{30} \Rightarrow x_4 = -\frac{1}{25}$$

$$\frac{3\sqrt{22}}{11}x_3 + \frac{\sqrt{22}}{6}x_4 = \frac{\sqrt{22}}{66} \Rightarrow x_3 = \frac{2}{25}$$

$$\frac{\sqrt{11}}{2}x_2 + \frac{2\sqrt{11}}{11}x_3 = -\frac{\sqrt{11}}{22} \Rightarrow x_2 = -\frac{3}{25}$$

$$2x_1 + \frac{1}{2}x_2 = \frac{1}{2} \Rightarrow x_1 = \frac{7}{25}$$

$$\mathbf{x}_1 = \begin{bmatrix} \frac{7}{25} \\ -\frac{3}{25} \\ \frac{2}{25} \\ -\frac{1}{25} \end{bmatrix}$$

(b) $\mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

Forward substitution: $\mathbf{L}\hat{\mathbf{b}} = \mathbf{b}_2$

Let:

$$\hat{\mathbf{b}} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

Then:

$$2y_1 = 0 \;\Rightarrow\; y_1 = 0$$

$$\frac{1}{2}y_1 + \frac{\sqrt{11}}{2}y_2 = 1 \;\Rightarrow\; y_2 = \frac{2}{\sqrt{11}} = \frac{2\sqrt{11}}{11}$$

$$\frac{2\sqrt{11}}{11}y_2 + \frac{3\sqrt{22}}{11}y_3 = 0 \;\Rightarrow\; y_3 = -\frac{2\sqrt{22}}{33}$$

$$\frac{\sqrt{22}}{6}y_3 + \frac{5\sqrt{2}}{6}y_4 = 0 \;\Rightarrow\; y_4 = \frac{2\sqrt{2}}{15}$$

$$\hat{\mathbf{b}}_2 = \begin{bmatrix} 0 \\ \frac{2\sqrt{11}}{11} \\ -\frac{2\sqrt{22}}{33} \\ \frac{2\sqrt{2}}{15} \end{bmatrix}$$

Back substitution: $\mathbf{L}^T\mathbf{x} = \hat{\mathbf{b}}_2$

$$\frac{5\sqrt{2}}{6}x_4 = \frac{2\sqrt{2}}{15} \;\Rightarrow\; x_4 = \frac{4}{25}$$

$$\frac{3\sqrt{22}}{11}x_3 + \frac{\sqrt{22}}{6}x_4 = -\frac{2\sqrt{22}}{33} \;\Rightarrow\; x_3 = -\frac{8}{25}$$

$$\frac{\sqrt{11}}{2}x_2 + \frac{2\sqrt{11}}{11}x_3 = \frac{2\sqrt{11}}{11} \;\Rightarrow\; x_2 = \frac{12}{25}$$

$$2x_1 + \frac{1}{2}x_2 = 0 \;\Rightarrow\; x_1 = -\frac{3}{25}$$

$$\mathbf{x}_2 = \begin{bmatrix} -\frac{3}{25} \\ \frac{12}{25} \\ -\frac{8}{25} \\ \frac{4}{25} \end{bmatrix}$$

(c) $\mathbf{b}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

Forward substitution: $\mathbf{L}\hat{\mathbf{b}} = \mathbf{b}_3$

Let:

$$\hat{\mathbf{b}} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

Then:

$$2y_1 = 1 \;\Rightarrow\; y_1 = \frac{1}{2}$$

$$\frac{1}{2}y_1 + \frac{\sqrt{11}}{2}y_2 = 1 \;\Rightarrow\; y_2 = \frac{3}{2\sqrt{11}} = \frac{3\sqrt{11}}{22}$$

$$\frac{2\sqrt{11}}{11}y_2 + \frac{3\sqrt{22}}{11}y_3 = 1 \;\Rightarrow\; y_3 = \frac{4\sqrt{22}}{33}$$

$$\frac{\sqrt{22}}{6}y_3 + \frac{5\sqrt{2}}{6}y_4 = 1 \;\Rightarrow\; y_4 = \frac{\sqrt{2}}{3}$$

$$\hat{\mathbf{b}}_3 = \begin{bmatrix} \frac{1}{2} \\ \frac{3\sqrt{11}}{22} \\ \frac{4\sqrt{22}}{33} \\ \frac{\sqrt{2}}{3} \end{bmatrix}$$

Back substitution: $\mathbf{L}^T\mathbf{x} = \hat{\mathbf{b}}_3$

$$\frac{5\sqrt{2}}{6}x_4 = \frac{\sqrt{2}}{3} \;\Rightarrow\; x_4 = \frac{2}{5}$$

$$\frac{3\sqrt{22}}{11}x_3 + \frac{\sqrt{22}}{6}x_4 = \frac{4\sqrt{22}}{33} \;\Rightarrow\; x_3 = \frac{1}{5}$$

$$\frac{\sqrt{11}}{2}x_2 + \frac{2\sqrt{11}}{11}x_3 = \frac{3\sqrt{11}}{22} \;\Rightarrow\; x_2 = \frac{1}{5}$$

$$2x_1 + \frac{1}{2}x_2 = \frac{1}{2} \;\Rightarrow\; x_1 = \frac{1}{5}$$

$$\mathbf{x}_3 = \begin{bmatrix} \frac{1}{5} \\ \frac{1}{5} \\ \frac{1}{5} \\ \frac{2}{5} \end{bmatrix}$$

Question 8 — Point Jacobi Iterations (20 Points)

Using the SPD system from Question 7 and $\mathbf{b}_1$:

1. Write the Point Jacobi update formula for the 4 unknowns.
2. Perform **two Jacobi iterations** starting from

$$\mathbf{x}^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

3. Perform **two Jacobi iterations** starting from

$$\mathbf{x}^{(0)} = \frac{1}{2}\mathbf{x}^* = \frac{1}{2}\begin{bmatrix} x_1^* \\ x_2^* \\ x_3^* \\ x_4^* \end{bmatrix}$$

   where $\mathbf{x}^*$ is your solution from Question 7 for $\mathbf{b}_1$.
4. For each starting guess, report the residual norm

$$\|\mathbf{K}\mathbf{x}^{(m)} - \mathbf{b}^{(1)}\|_2$$

   after each iteration ($m = 1$ and $m = 2$) (you may approximate to 2–3 decimals).

Solution for Question8

1. Write the Point Jacobi update formula for the 4 unknowns.

$$x_1^{(m)} = \frac{b_1 - a_{12}x_2^{(m-1)} - a_{13}x_3^{(m-1)} - a_{14}x_4^{(m-1)}}{a_{11}}$$

$$x_2^{(m)} = \frac{b_2 - a_{21}x_1^{(m-1)} - a_{23}x_3^{(m-1)} - a_{24}x_4^{(m-1)}}{a_{22}}$$

$$x_3^{(m)} = \frac{b_3 - a_{31}x_1^{(m-1)} - a_{32}x_2^{(m-1)} - a_{34}x_4^{(m-1)}}{a_{33}}$$

$$x_4^{(m)} = \frac{b_4 - a_{41}x_1^{(m-1)} - a_{42}x_2^{(m-1)} - a_{43}x_3^{(m-1)}}{a_{44}}$$

2. Perform **two Jacobi iterations** starting from

$$\mathbf{x}^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- **First Iteration**

$$x_1^{(1)} = \frac{b_1 - a_{12}x_2^{(0)} - a_{13}x_3^{(0)} - a_{14}x_4^{(0)}}{a_{11}} = \frac{1-0-0-0}{4} = \frac{1}{4}$$

$$x_2^{(1)} = \frac{b_2 - a_{21}x_1^{(0)} - a_{23}x_3^{(0)} - a_{24}x_4^{(0)}}{a_{22}} = \frac{0-0-0-0}{3} = 0$$

$$x_3^{(1)} = \frac{b_3 - a_{31}x_1^{(0)} - a_{32}x_2^{(0)} - a_{34}x_4^{(0)}}{a_{33}} = \frac{0-0-0-0}{2} = 0$$

$$x_4^{(1)} = \frac{b_4 - a_{41}x_1^{(0)} - a_{42}x_2^{(0)} - a_{43}x_3^{(0)}}{a_{44}} = \frac{0-0-0-0}{2} = 0$$

$$\mathbf{x}^{(1)} = \begin{bmatrix} \frac{1}{4} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- **Second Iteration**

$$x_1^{(2)} = \frac{b_1 - a_{12}x_2^{(1)} - a_{13}x_3^{(1)} - a_{14}x_4^{(1)}}{a_{11}} = \frac{1-0-0-0}{4} = \frac{1}{4}$$

$$x_2^{(2)} = \frac{b_2 - a_{21}x_1^{(1)} - a_{23}x_3^{(1)} - a_{24}x_4^{(1)}}{a_{22}} = \frac{0-\frac{1}{4}-0-0}{3} = -\frac{1}{12}$$

$$x_3^{(2)} = \frac{b_3 - a_{31}x_1^{(1)} - a_{32}x_2^{(1)} - a_{34}x_4^{(1)}}{a_{33}} = \frac{0-0-0-0}{2} = 0$$

$$x_4^{(2)} = \frac{b_4 - a_{41}x_1^{(1)} - a_{42}x_2^{(1)} - a_{43}x_3^{(1)}}{a_{44}} = \frac{0-0-0-0}{2} = 0$$

$$\mathbf{x}^{(2)} = \begin{bmatrix} \frac{1}{4} \\ -\frac{1}{12} \\ 0 \\ 0 \end{bmatrix}$$

3. Perform **two Jacobi iterations** starting from

$$\mathbf{x}^{(0)} = \frac{1}{2}\mathbf{x}^* = \frac{1}{2}\begin{bmatrix} x_1^* \\ x_2^* \\ x_3^* \\ x_4^* \end{bmatrix}$$

where $\mathbf{x}^*$ is your solution from Question 7 for $\mathbf{b}_1$.

$$\mathbf{x}^{(0)} = \frac{1}{2}\mathbf{x}^* = \frac{1}{2}\begin{bmatrix} x_1^* \\ x_2^* \\ x_3^* \\ x_4^* \end{bmatrix} = \mathbf{x}_1 = \begin{bmatrix} \frac{7}{50} \\ -\frac{3}{50} \\ \frac{2}{50} \\ -\frac{1}{50} \end{bmatrix}$$

- **First Iteration**

$$x_1^{(1)} = \frac{b_1 - a_{12}x_2^{(0)} - a_{13}x_3^{(0)} - a_{14}x_4^{(0)}}{a_{11}} = \frac{1 - (-\frac{3}{50}) - 0 - 0}{4} = \frac{53}{200}$$

$$x_2^{(1)} = \frac{b_2 - a_{21}x_1^{(0)} - a_{23}x_3^{(0)} - a_{24}x_4^{(0)}}{a_{22}} = \frac{0 - \frac{7}{50} - \frac{1}{25} - 0}{3} = -\frac{3}{50}$$

$$x_3^{(1)} = \frac{b_3 - a_{31}x_1^{(0)} - a_{32}x_2^{(0)} - a_{34}x_4^{(0)}}{a_{33}} = \frac{0 - (-\frac{3}{50}) - (-\frac{1}{50}) - 0}{2} = \frac{1}{25}$$

$$x_4^{(1)} = \frac{b_4 - a_{41}x_1^{(0)} - a_{42}x_2^{(0)} - a_{43}x_3^{(0)}}{a_{44}} = \frac{0 - 0 - 0 - \frac{1}{25}}{2} = -\frac{1}{50}$$

$$\mathbf{x}^{(1)} = \begin{bmatrix} \frac{53}{200} \\ -\frac{3}{50} \\ \frac{1}{25} \\ -\frac{1}{50} \end{bmatrix}$$

- **Second Iteration**

$$x_1^{(2)} = \frac{b_1 - a_{12}x_2^{(1)} - a_{13}x_3^{(1)} - a_{14}x_4^{(1)}}{a_{11}} = \frac{1 - (-\frac{3}{50}) - 0 - 0}{4} = \frac{53}{200}$$

$$x_2^{(2)} = \frac{b_2 - a_{21}x_1^{(1)} - a_{23}x_3^{(1)} - a_{24}x_4^{(1)}}{a_{22}} = \frac{0 - \frac{53}{200} - \frac{1}{25} - 0}{3} = -\frac{61}{600}$$

$$x_3^{(2)} = \frac{b_3 - a_{31}x_1^{(1)} - a_{32}x_2^{(1)} - a_{34}x_4^{(1)}}{a_{33}} = \frac{0 - (-\frac{3}{50}) - (-\frac{1}{50}) - 0}{2} = \frac{1}{25}$$

$$x_4^{(2)} = \frac{b_4 - a_{41}x_1^{(1)} - a_{42}x_2^{(1)} - a_{43}x_3^{(1)}}{a_{44}} = \frac{0 - 0 - 0 - \frac{1}{25}}{2} = -\frac{1}{50}$$

$$\mathbf{x}^{(2)} = \begin{bmatrix} \frac{53}{200} \\ -\frac{61}{600} \\ \frac{1}{25} \\ -\frac{1}{50} \end{bmatrix}$$

4. For each starting guess, report the residual norm

$$\|\mathbf{K}\mathbf{x}^{(m)} - \mathbf{b}^{(1)}\|_2$$

after each iteration ($m = 1$ and $m = 2$) (you may approximate to 2–3 decimals).

4. Residual norms for each starting guess (using results from Q2 and Q3)

We report the Euclidean residual norm:

$$\|Kx^{(m)} - b^{(1)}\|_2,$$

where

$$K = \begin{bmatrix} 4 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}, \qquad b^{(1)} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

- **Case A (from Q2): start with** $x^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

Residual at $m = 1$:

$$Kx^{(1)} = \begin{bmatrix} 4 \times \frac{1}{4} + 1 \times 0 \\ 1 \times \frac{1}{4} + 3 \times 0 + 1 \times 0 \\ 1 \times 0 + 2 \times 0 + 1 \times 0 \\ 1 \times 0 + 2 \times 0 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{4} \\ 0 \\ 0 \end{bmatrix}$$

$$r^{(1)} = Kx^{(1)} - b^{(1)} = \begin{bmatrix} 0 \\ \frac{1}{4} \\ 0 \\ 0 \end{bmatrix}$$

$$\|r^{(1)}\|_2 = \frac{1}{4}$$

Residual at $m = 2$:

$$Kx^{(2)} = \begin{bmatrix} 4 \times \frac{1}{4} + 1 \times \left(-\frac{1}{12}\right) \\ 1 \times \frac{1}{4} + 3 \times \left(-\frac{1}{12}\right) + 1 \times 0 \\ 1 \times \left(-\frac{1}{12}\right) + 2 \times 0 + 1 \times 0 \\ 1 \times 0 + 2 \times 0 \end{bmatrix} = \begin{bmatrix} \frac{11}{12} \\ 0 \\ -\frac{1}{12} \\ 0 \end{bmatrix}$$

$$r^{(2)} = Kx^{(2)} - b^{(1)} = \begin{bmatrix} -\frac{1}{12} \\ 0 \\ -\frac{1}{12} \\ 0 \end{bmatrix}$$

$$\|r^{(2)}\|_2 = \sqrt{\left(\frac{1}{12}\right)^2 + \left(\frac{1}{12}\right)^2} = \frac{\sqrt{2}}{12}$$

- **Case B (from Q3): start with** $x^{(0)} = \frac{1}{2}x^*$

From Q3:

$$x^{(1)} = \begin{bmatrix} \frac{53}{200} \\ -\frac{3}{50} \\ \frac{1}{25} \\ -\frac{1}{50} \end{bmatrix}, \qquad x^{(2)} = \begin{bmatrix} \frac{53}{200} \\ -\frac{61}{600} \\ \frac{1}{25} \\ -\frac{1}{50} \end{bmatrix}$$

Residual at $m = 1$:

$$Kx^{(1)} = \begin{bmatrix} 4 \times \frac{53}{200} + 1 \times \left(-\frac{3}{50}\right) \\ 1 \times \frac{53}{200} + 3 \times \left(-\frac{3}{50}\right) + 1 \times \frac{1}{25} \\ 1 \times \left(-\frac{3}{50}\right) + 2 \times \frac{1}{25} + 1 \times \left(-\frac{1}{50}\right) \\ 1 \times \frac{1}{25} + 2 \times \left(-\frac{1}{50}\right) \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{8} \\ 0 \\ 0 \end{bmatrix}$$

$$r^{(1)} = Kx^{(1)} - b^{(1)} = \begin{bmatrix} 0 \\ \frac{1}{8} \\ 0 \\ 0 \end{bmatrix}, \qquad \|r^{(1)}\|_2 = \frac{1}{8}$$

Residual at $m = 2$:

$$Kx^{(2)} = \begin{bmatrix} 4 \times \frac{53}{200} + 1 \times \left(-\frac{61}{600}\right) \\ 1 \times \frac{53}{200} + 3 \times \left(-\frac{61}{600}\right) + 1 \times \frac{1}{25} \\ 1 \times \left(-\frac{61}{600}\right) + 2 \times \frac{1}{25} + 1 \times \left(-\frac{1}{50}\right) \\ 1 \times \frac{1}{25} + 2 \times \left(-\frac{1}{50}\right) \end{bmatrix} = \begin{bmatrix} \frac{575}{600} \\ 0 \\ -\frac{25}{600} \\ 0 \end{bmatrix}$$

$$r^{(2)} = Kx^{(2)} - b^{(1)} = \begin{bmatrix} -\frac{25}{600} \\ 0 \\ -\frac{25}{600} \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{24} \\ 0 \\ -\frac{1}{24} \\ 0 \end{bmatrix}$$

$$\|r^{(2)}\|_2 = \sqrt{\left(\frac{1}{24}\right)^2 + \left(\frac{1}{24}\right)^2} = \frac{\sqrt{2}}{24}$$

- Summary

Start with $x^{(0)} = \mathbf{0}$:

$$\|Kx^{(1)} - b^{(1)}\|_2 = \frac{1}{4}, \qquad \|Kx^{(2)} - b^{(1)}\|_2 = \frac{\sqrt{2}}{12}.$$

Start with $x^{(0)} = \frac{1}{2}x^*$:

$$\|Kx^{(1)} - b^{(1)}\|_2 = \frac{1}{8}, \qquad \|Kx^{(2)} - b^{(1)}\|_2 = \frac{\sqrt{2}}{24}.$$